REDROB DATA & AI CHALLENGE · HACK2SKILL · 2026

Intelligent Candidate Discovery & Ranking

A Two-Phase Hybrid Ranking System for 100,000 Profiles

100K	**Top 100**	**23.3s**	**0**
Candidate Profiles	Best-Fit Output	Ranking Runtime	Honeypots in Top 100

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The Challenge

What makes this harder than it looks

⚡ Scale & Constraints

Dataset:	100,000 synthetic candidate profiles
Target:	Senior AI Engineer — Founding Team
Output:	Top 100 best-fit candidates as CSV
Runtime:	≤ 5 min wall-clock, CPU only
Memory:	≤ 16 GB RAM
Network:	Offline — no API calls during ranking
GPU:	Not allowed during ranking step

⚠ The Keyword-Stuffing Trap

The trap:	Naive embedding/keyword match ranks buzzword-stuffers highly
Example:	Marketing Manager listing 10 AI tools outranks a real ML engineer
Root cause:	Skills list ≠ actual production experience
Dataset design:	~80 honeypot profiles with impossible data planted deliberately
Disqualifier:	>10% honeypots in top 100 = full submission disqualified
Our answer:	Hand-engineered hybrid scorer + title taxonomy + duration-weighted skills

🔬 No Training Signal

Ground truth:	Hidden — revealed only after submissions close
Implication:	Cannot train a supervised ranking model
Approach:	Transparent, explainable, hand-tuned hybrid scorer
Calibration:	Hand-labeled 50-candidate sample as proxy evaluation set
Evaluation:	NDCG@10 (50%) + NDCG@50 (30%) + MAP (15%) + P@10 (5%)
Stage 5:	30-min interview to defend architecture decisions

Two-Phase Architecture

The key design decision that meets all hard constraints

⏸ PHASE 1 — OFFLINE (no time limit)

precompute_features.py

- Parses all 100K candidates from candidates.jsonl
- Extracts 8 rule-based scoring features per candidate
- Runs 4 honeypot consistency checks — flags impossible profiles
- Output: artifacts/features.pkl.gz (~31 MB compressed)

precompute_embeddings.py

- Loads all-MiniLM-L6-v2 model locally (80 MB, CPU-only)
- Encodes headline + summary + career descriptions per candidate
- Truncated to ~300 words per profile for encoding speed
- Output: embeddings.npy — exactly 100K × 384 float32 = 153 MB
- Also encodes the fixed JD → jd_embedding.npy

▶ PHASE 2 — RANKING (under 25 seconds)

rank.py (the timed step)

- Loads features.pkl.gz + embeddings.npy from disk instantly
- Cosine similarity: one np.dot(embeddings, jd_vec) — milliseconds
- Applies hybrid score formula across all 99,939 valid candidates
- Excludes 61 honeypot-flagged candidates from top-100 eligibility
- Sorts by final_score descending, tie-break: candidate_id ascending

Reasoning generator (inside rank.py)

- Pulls specific facts: years, skill name, duration_months, location
- Selects among 6 sentence templates based on dominant score driver
- Tone varies by rank: confident at #1, hedged/honest at #100
- Output: submission.csv — 100 rows, all fact-grounded

Hybrid Scoring Formula

score.py — all weights are named constants, fully adjustable

Component	Weight	Details
Title Relevance	25%	Custom taxonomy: ML/AI Eng=1.0 · Backend(ML)=0.6 · Marketing Mgr=0.0 · Civil Eng=0.0
Production ML Evidence	20%	Scans career_history descriptions for 'deployed','shipped','production','real users' — boosted at product companies
Retrieval / Infra Skills	15%	Pinecone, Weaviate, FAISS, OpenSearch, BGE, E5, sentence-transformers — weighted by duration_months × endorsements
Semantic Similarity	15%	Cosine similarity of candidate embedding vs JD embedding — supports but never dominates the decision
Evaluation Framework	10%	NDCG, MRR, MAP, A/B testing mentions in skills and career history — weighted by duration/endorsements
Experience Years Fit	5%	Smooth bump function peaking at 5-9 years — decays gradually outside range, never a hard cutoff
Location Fit	5%	Pune/Noida=highest · Other Tier-1 + willing_to_relocate=good · Outside India=low
Notice Period	5%	Smooth penalty scaling above 30 days — 90-day notice visibly hurts but doesn't zero-out

final_score = **base_fit_score** × **behavioral_multiplier** [0.70 – 1.15] | Signals: last_active_date · open_to_work · recruiter_response_rate · interview_completion_rate / *−1 sentinel → neutral*

Honeypot Consistency Gate

Candidates failing ANY check are excluded entirely — not just down-ranked

01 Unearned Expertise

- Expert proficiency claimed in a skill with 0 duration_months
- Physically impossible — you cannot be an expert at something
- you have never actually used
- Catches: candidates inflating skill levels with no evidence

02 Experience Mismatch

- Stated years_of_experience wildly inconsistent with the
- sum of all career_history job durations
- Generous slack applied for employment gaps between roles
- Catches: profiles claiming 15 yrs exp with 2 yrs of history

03 Impossible Timeline

- Any job where start_date is after its own end_date
- A physically impossible employment record
- No reasonable interpretation makes this legitimate
- Catches: data-generation errors seeded as deliberate traps

04 Tenure Discrepancy

- duration_months field contradicts actual start/end date gap
- by more than 6 months — generous threshold to avoid false positives
- Validated: gap between stated and calculated must be egregious
- Catches: the most subtle honeypot type in the dataset

✓ **Result:** 61 true honeypots excluded (0.06% of dataset) | Key investigation: Initially 24,931 flagged — salary-inversion & signup-date rules were random data noise. Proved via organiser's own sample file. Removed.

Real Engineering: Catching the Keyword-Stuffing Trap

A concrete example of catching the exact failure mode the dataset is designed to test for

✕ BEFORE — broken

Root cause:
title_relevance_score was not discriminating between AI/ML roles and completely unrelated titles

What happened:

Marketing Manager (CAND_0000039) ranked #9 out of 50 candidates

Reasoning text:

"Exceptional ML engineer profile with 3.9 years of background..."

Other examples:
Civil Engineer ranked #13, Project Manager #5 — all praised as 'Highly relevant'

Root cause:
Taxonomy keyword list was too permissive — matching on generic 'Engineer' rather than AI/ML-specific terms

Impact at scale:
If wrong on 50-candidate sample, identical bug running on full 100K submission CSV

→

✓ AFTER — fixed

Fix applied:
Rebuilt title taxonomy with explicit zero-scoring for all unrelated title categories

Taxonomy:
ML/AI Engineer → 1.0 · Backend Eng (ML) → 0.6 · Marketing Mgr → 0.0 · Civil Eng → 0.0

CAND_0000039:

Marketing Manager now ranked #10/50

New reasoning:
"Background as Marketing Manager is not aligned with the core requirements"

Scale check:

Full 50-row scan for consecutive duplicate words — 0 matches after fix

100K rerun:
Re-ran full pipeline: all top-100 titles are AI/ML-relevant, none are unrelated roles

Verified Results

Every number below was measured on the real 100,000-candidate dataset

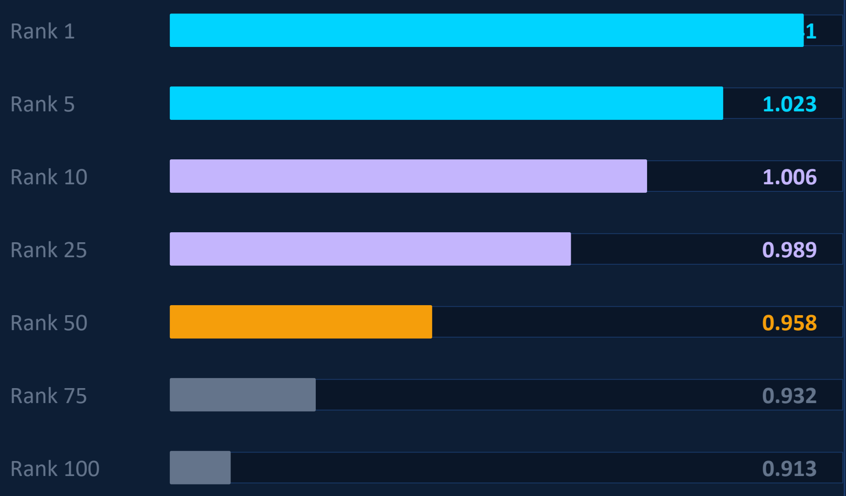
23.3s
Phase 2 wall-clock
(5-min budget = 13× headroom)

61
Honeypots excluded
(0 in top 100)

VALID
validate_submission.py
zero format errors

19
Git commits
real incremental history

Score Distribution



Verification Details

Hardware: 13th Gen Intel i5-13500H · 16 GB RAM

Runtime: 23.33s Phas-e 2 · ~6 hrs Phase 1 (once only)

Network: Fully offline during ranking — zero API calls

GPU: Not used at any point in the pipeline

Tie-break: Deterministic: candidate_id ascending

Score range: 1.041 (rank 1) → 0.913 (rank 100)

Reasoning: All 100 rows verified — zero duplication patterns

Git history: 19 commits: scaffold→features→scoring→tests→Streamlit→bug fixes→docs

Summary

01

Two-Phase Architecture

Embedding precompute runs once offline (no time limit). Ranking step loads arrays and runs matrix math — 23.3s, no API, no GPU, no network.

02

Hybrid Scoring Defeats Keyword Stuffing

Custom title taxonomy (Marketing Mgr=0.0) + duration-weighted skills + production evidence stops buzzword-stuffers at the title-relevance gate before semantic similarity even runs.

03

Evidence-Based Honeypot Gate

Four structural consistency checks. Validated against organiser's own sample: proved salary-inversion rule was random noise affecting 25% of candidates. 61 true honeypots excluded.

04

Real Bug Caught and Fixed

Marketing Manager ranked #9/50 with reasoning praising them as 'Exceptional ML engineer.' Diagnosed, title taxonomy rebuilt, re-ran full 100K — zero unrelated titles in top 100.

GitHub: https://github.com/midhunpm1002/redrob-candidate-ranking | Sandbox: https://redrob-candidate-ranking.streamlit.app